

Item	Weak #1	Basic #2	Average #3	Proficient #4	Advanced #5
Assessment (8 marks)					
8 marks	1*	1*	1*	2*	3
(A) Quality of the self-assessment (individual)	Most items are not as claimed and contain unreported major or multiple minor issues	Most items are as claimed but a few major issues are not reported	Almost all items are as claimed except for some minor issues	Proposed marks are coherent with the comments and marking scheme	All items are as claimed
One page comments (9 marks)					
3 marks each		1	1	1	
(B) For each of the 3 issues or additional feature commented		The issue or new feature is briefly explained and relevant to networks	The comment mentions the changes needed in the protocol model or processing	The comment adequately mentions the corresponding part of the IRC protocol	
Common (14 marks)					
7 marks	1*	1*	1*	2	2
(C) Modelling all the necessary parts of the IRC protocol¹	Some of the data-structure required to model the protocol are not defined	All needed data-structures are defined but some are incomplete (missing properties or functions)	All necessary data-structures of the protocol are correctly modelled and used where appropriate	The parsing of IRC messages is efficient and isolated from network operations	Generation of responses is isolated from network operations ² and parsing of messages
7 marks	1	1	1	2	2
(D) Source code documentation, readability and software development	Appropriate use of variable and function names and spacing to make the code reading intuitive	All relevant code (variables, functions and complex pieces), and only them, are commented	Comments are clear and concise, easily readable	Comments focus on explaining why the code do this and not describing how	Functions are kept relatively short and defined for every redundant code
<i>* to be assessed in mid-submission</i>					

¹ Data-structure is intended as classes with appropriate methods in Python or C++. In C, this is the data-structure themselves and the functions to deal with them.

² Network operations refers to all socket related calls such as bind, recv, send,...

Item	Weak #1	Basic #2	Average #3	Proficient #4	Advanced #5
Bot (24 marks)					
7 marks	1*	1*	1*	2*	2
(E) The bot can connect to a miniircd server, handle errors and stay connected	Bot instantiates data-structures required for network communication	Bot establishes the connection with the server using the correct VM address	Bot correctly identify himself to the miniircd server	Keeping the connection to miniircd alive, and managing it efficiently ³	Reacting to common connection errors or network issues
4 marks		1*	1*	2	
(F) The bot can join a channel, retain the user list and keep track of users		Bot correctly request to join the channel and hexchat can display it joining the channel	Retaining the initial information sent by miniircd about the channel and its users	Updating information about the channel users efficiently when they join or leave	
5 marks		1	1	1	2
(G) The bot responds automatically to private messages with random facts or joke delivered privately		Bot correctly receive incoming private messages from miniircd	Bot correctly process incoming private messages from miniircd	The sender receive a random fact in private and hexchat can display it properly in the correct private tab	Random facts are stored in a local file or retrieved from the web
5 marks	1	1	2	1	
(H) The bot responds to bot commands on the channel and hexchat can display it correctly in the channel tab	Bot correctly receive and process incoming channel messages from miniircd	Responding correctly to the !hello command and hexchat can display it properly in the correct channel tab	Responding correctly to the !slap command and hexchat can display it properly in the correct channel tab	The !slap command supports the optional parameter and ensures the slapped user is neither the bot nor the sender	
3 marks		1	1	1	
(I) Extra bot feature (private ⁴ or channel related), tested only on miniircd		Uses same IRC commands similarly to previous items	Uses one new IRC command not used in previous items with similar purpose	Uses many IRC commands not used in previous items or different purpose	
* to be assessed in mid-submission					

³ This means not using all resource of the system (average CPU load stays low, i.e. < 10%)

⁴ Private here does not mean private message, it can be anything involving only 2 clients.

Item	Weak #1	Basic #2	Average #3	Proficient #4	Advanced #5
Server (45 marks)					
9 marks	1*	2*	2*	2	2
(J) Server allows connections from multiple IRC compliant clients like hexchat	Server instantiates data-structures required for network communication	Server allows and remembers a new hexchat client connection	Server retain the login information provided by an hexchat client and welcomes them	Server can handle multiple hexchat clients without ambiguity	Server manage multiple hexchat clients efficiently for the system ⁵
9 marks	1*	2 (1)*	2 (1)*	2	2
(K) Server correctly reacts to errors or unusual circumstances	Server properly handle unknown commands ⁶ and hexchat shows the error correctly	Server detects invalid nickname and informs hexchat correctly so it can be changed automatically	Server does not crash, show warnings or send garbage data to users under normal traffic	Server reacts properly to incorrectly formatted messages or parameters errors and hexchat shows it	Server detects disconnected or unresponsive ⁷ clients and removes them adequately
7 marks	1	1	2	1	2
(L) Users with an IRC compliant clients such as hexchat can talk privately	Server correctly receive incoming private messages from hexchat clients	Server correctly process incoming private messages from hexchat clients	Destination user can display received messages correctly in in the appropriate tab	Users only receive messages they legitimately should	Server handles most common errors related to private messages
10 marks	1	2	2	3	2
(M) Users with an IRC compliant clients such as hexchat can join and leave channels	Server correctly receives requests to join channels from hexchat clients	Server correctly processes requests to join channels and informs hexchat back	Users already in the channel display the new user arrival in hexchat	New user displays the list of users connected to the channel in the correct hexchat tab	Users can leave a channel or the server and other users display it correctly
10 marks	1	2	3	2	2
(N) Users with an IRC compliant clients such as hexchat can talk on channels	Server correctly receive incoming channel messages	Server correctly process incoming channel messages	Users in the channel display all messages from other users properly	Users only receive messages they should	Server handles most common errors related to channel messages
* to be assessed in mid-submission (K2 and K3 partial)					

⁵ This means not using all resource of the system (average CPU load stays low, i.e. < 10%).

⁶ Commands which are not implemented due to being out of the scope of this assignment are considered unknown even if they exist in the protocol.

⁷ Unresponsive clients are defined as client being idle (i.e. no command received) more than 1 minute.